

Summer Project 2

Rebuilding Team 2930's 2023

robot code

Why Team 2930?

I chose the 2023 Sonic Squirrels robot because it was a complete competition codebase to learn from. Charged Up gave me real drive, scoring, intake, and vision problems to study. I kept that behavior while rebuilding the structure around IO-based devices.

The code structure

WPILib's RobotContainer connects controllers to commands. Each subsystem owns one job, while IO classes separate real hardware from simulation. Motor, sensor, and vision inputs are logged through AdvantageKit, which makes testing and debugging easier.

Drivetrain

Four-module swerve drive uses TalonFX drive motors, TalonFX steer motors, CANcoders, and a Pigeon gyro. The 2023 geometry, CAN IDs, gear ratios, and wheel offsets are carried over. Field-relative control, acceleration limiting, X-stance, odometry, and pose estimation round out the drivetrain.

Elevator, stinger, and intake

Elevator — two Falcon 500s run as a follower pair with Motion Magic position control, current limiting, and software height limits.

Stinger — one Falcon 500 controls extension with position targets, retraction protection, and an elevator-clearance check.

Intake — one motor switches between cone and cube directions with separate intake, hold, and eject states.

All three expose simple commands through RobotContainer.

Vision and LEDs

Vision — three PhotonVision cameras feed pose estimates using the 2023 Charged Up AprilTag layout. New tagged measurements are sent to the drivetrain pose estimator, with camera transforms and a reference pose used to reject stale data.

LEDs — the REV Blinkin controller uses PWM patterns for quick status feedback during testing.

What I learned

Built from the outside in: map the hardware, define each subsystem's job, then connect controls and autonomous actions. The code compiles and its unit tests pass, but real-bot checks still matter — especially swerve offsets, mechanism limits, and camera names.